Text Classification Project Report

1. Objectives

The main goal of the project was to study how different text feature representations, feature selection methods, and classification algorithms affect sentiment classification performance on the NLTK Movie Review dataset.

Specific experimental goals were:

- Compare term frequency vs TF-IDF representations, stop-word removal, and unigram/bigram
- Evaluate two feature selection strategies:
    - Frequency-based filtering
    - Mutual Information
- Compare three classifier models
    - Multinomial Naïve-Bayes
    - Logistic Regression
    - SVM
- Explore an ensemble model and determine if it outperforms other models
- Identify the best-performing, most accurate models and processes

2. Data Description

2.1 Dataset Source

The project used the Movie Reviews Dataset from the NLTK library. It contains 1,000 negative reviews and 1,000 positive reviews for movie sentiment classification.

2.2 Dataset Characteristics

- Total documents: 2,000
- Positive reviews: 1,000
- Negative reviews: 1,000
- Balanced class distribution

The dataset is balanced, so accuracy was selected as the main evaluation metric.

2.3 Preprocessing Impact

Before preprocessing:

- Text documents with raw tokenization

After vectorization:

- Sparse feature matrices

- Dimensionality differed based on:
    - Unigrams vs. bigrams
    - Thresholds for feature filtering
    - Selection of mutual information

Example dimensional changes:

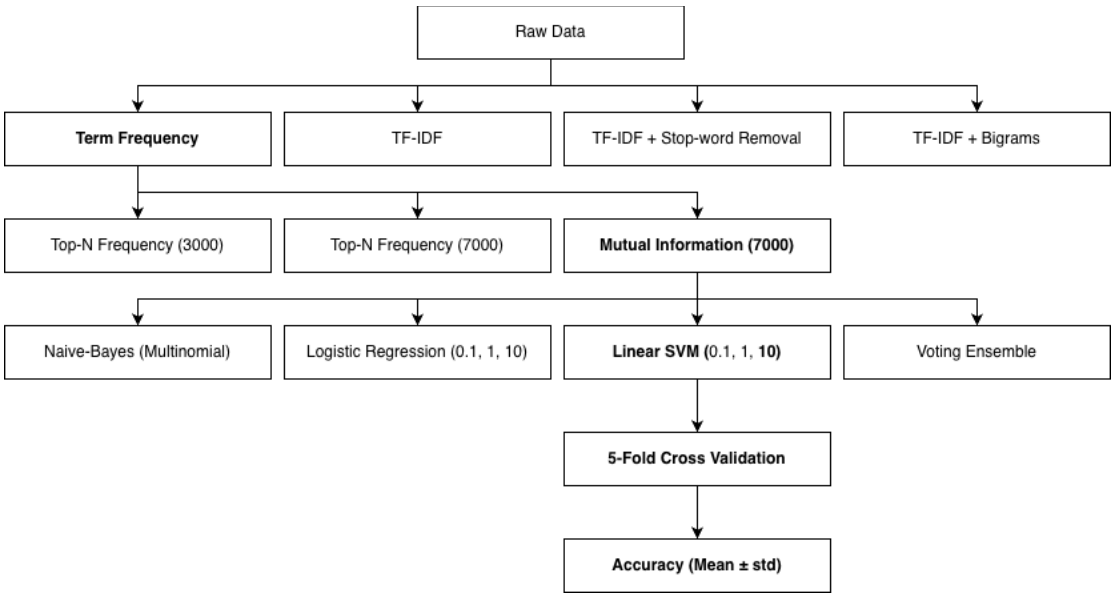
- Full TF-IDF vocabulary: More than 30,000 terms
- Top 7000 feature filtering: 7000 terms
- MI (Top 7000): 7000 selected discriminative terms

[Instructor note: Include dataset size, class distribution, and before/after preprocessing statistics.]

3. Experimental Pipeline Overview

3.1 Pipeline Flowchart

[Instructor note: A diagram is required. Clearly label alternatives tested at each stage.]



4. Feature Selection Methods

4.1 Method 1: Frequency-Based Filtering

The top *N* most common terms were selected:

- Top 3000
- Top 7000

Results:

Features	Accuracy
Top 3000	0.8235 ± 0.0150
Top 7000	0.8210 ± 0.0163

Findings: Frequency filtering did not significantly improve accuracy.

4.2 Method 2: Mutual Information

Mutual Information used class labels to rank features according to dependencies.

The top 7000 MI-ranked features were selected.

Findings:

Accuracy = **0.8355 ± 0.0168**

This demonstrated that dependency-based ranking more accurately captures discriminative sentiment terms than frequency-based selection.

5. Text Preprocessing Methods

[Instructor note: Compare with/without each method under identical validation settings.]

Comparison under identical settings **(Logistic Regression, 5-fold CV)**:

Method	Accuracy
Term frequency	0.8345 ± 0.0206
TF-IDF	0.8240 ± 0.0139
TF-IDF + stop-words removed	0.8175 ± 0.0106
TF-IDF + unigrams + bigrams	0.8045 ± 0.0175

Key findings:

- Term frequency alone outperformed TF-IDF in terms of accuracy.
- Accuracy was marginally reduced by stop-word removal, most likely due to the removal of negative terms such as “not” and “never”.
- Because of the smaller dataset size and increased dimensionality, adding bigrams decreased accuracy.

6. Classification Models

6.1 Model 1: Multinomial Naive Bayes

Accuracy: **0.8115 ± 0.0137**

Strong independence assumptions probably contributed to the lower accuracy compared to linear models.

6.2 Model 2: Logistic Regression

C	Accuracy
0.1	0.7925 ± 0.0088
1	0.8240 ± 0.0139
10	0.8435 ± 0.0168

Higher C (less regularization) improved accuracy.

6.3 Model 3: Linear SVM

C	Accuracy
0.1	0.8215 ± 0.0133
1	0.8465 ± 0.0140
10	0.8500 ± 0.0092

Linear SVM with a C of 10 achieved the best accuracy overall.

7. Ensemble Method

7.1 Ensemble Architecture

I used a voting classifier that integrated:

- Logistic Regression
- Linear SVM
- Naive Bayes

Hard voting was used.

7.2 Design Rationale

The ensemble was designed to combine:

- Probabilistic modeling (NB)
- Margin-based learning (SVM)
- Regularized linear classification (LR)

Result:

Accuracy = **0.8410 ± 0.0166**

The ensemble was not more accurate than linear SVM, which was 85% accurate. This implies that the majority of discriminative patterns were already captured by linear SVM, leaving little complementary signal for the ensemble model to improve.

[Instructor note: Ensemble must outperform best non-ensemble model or provide justified explanation.]

8. Algorithms and Pseudocode

Mutual Information Feature Selection

Input: Document-term matrix X, labels y

For each feature f:

  Compute mutual information between f and y

Rank features by MI score

Select top k features

Return reduced feature matrix

Linear SVM Classification

Input: Feature matrix X, labels y

Initialize SVM with regularization parameter C

Train SVM on training folds

Evaluate on validation fold

Repeat for 5 folds

Compute mean accuracy and standard deviation

9. Validation and Evaluation

9.1 Validation Protocol

- 5-fold cross-validation
- Stratified splits
- All experiments used the same folds

9.2 Evaluation Metrics

Primary metric:

Accuracy (mean ± standard deviation)

Example:

Linear SVM (C=10):

0.8500 ± 0.0092

10. Results

10.1 Quantitative Results

Best overall model:

Linear SVM (C=10)

Accuracy: **85%**

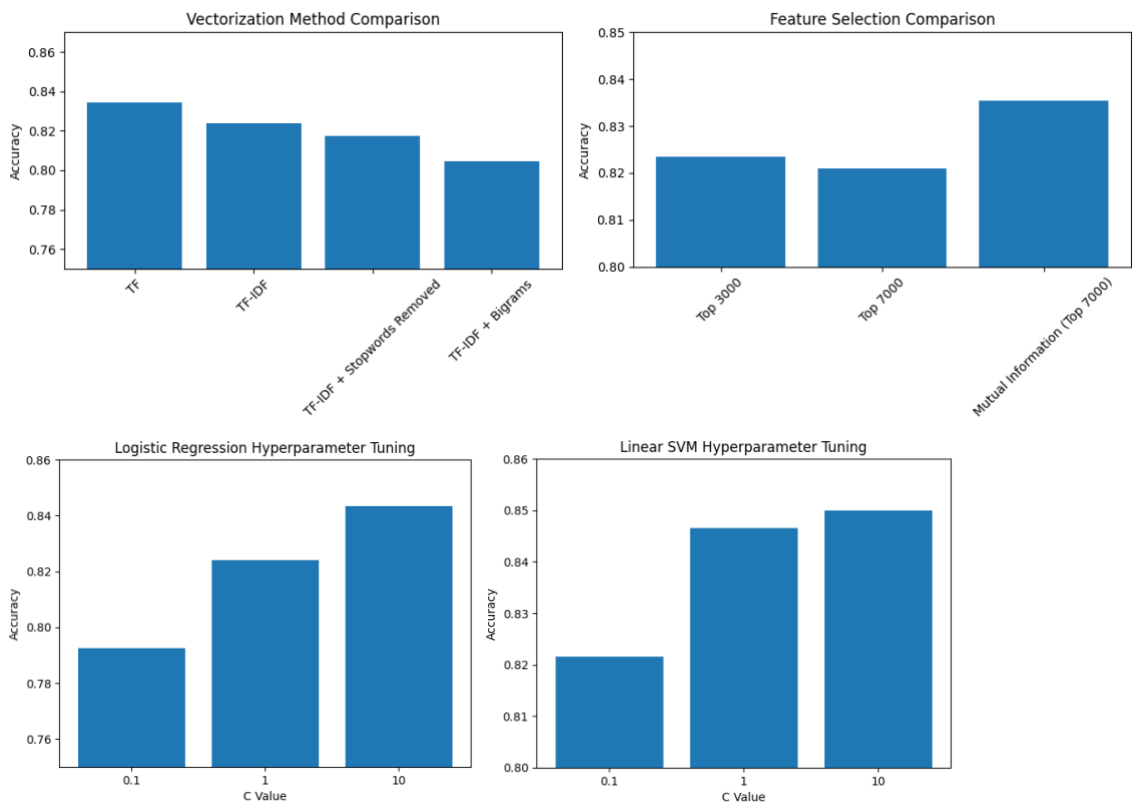
Best feature selection method:

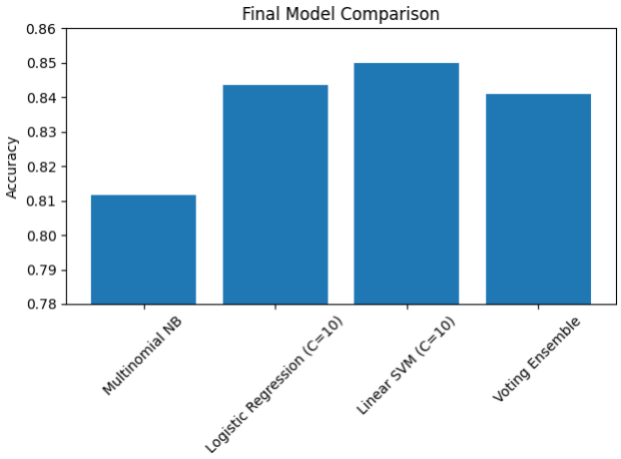
Mutual Information (Top 7000)

Best preprocessing technique:

Term frequency

10.2 Visualizations





11. Analysis and Discussion

Key trends:

1. Term frequency outperformed TF-IDF, suggesting that sentiment-bearing words benefit from raw frequency representation.
2. Stop-word removal slightly reduced performance, highlighting the importance of negation terms.
3. Mutual information improved performance over frequency filtering.
4. Linear SVM consistently outperformed Logistic Regression and Naive Bayes.
5. Increasing C improved performance, indicating mild underfitting at lower regularization strengths.
6. Ensemble did not outperform the best individual model, likely due to limited complementary diversity.

Trade-off:

Higher C improves accuracy but increases overfitting risk. However, low variance across folds suggests stable generalization.

12. Conclusion and Insights

The best-performing configuration combined:

- Term frequency representation
- Mutual information feature selection
- Linear SVM (C=10)

Achieving:

85.0% cross-validated accuracy

Key insight:

Feature representation and regularization strength had greater impact on performance than ensemble modeling.

Limitations:

- Small dataset size (2000 documents)
- Limited exploration of deep learning approaches
- No advanced preprocessing (lemmatization, embeddings)

Future Work:

- Apply transformer-based models
- Perform error analysis on misclassified reviews

References

NLTK

Movie Reviews Dataset

Pedregosa et al., 2011. Scikit-learn: Machine Learning in Python.